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SIMATS ENGINEERING

CO5 - ASSESSMENT TOOL 3

Technical Presentation

COURSE NAME: Deep Learning
SLOT : D

COURSE CODE: MLA0405
Faculty : Dr.Arul Raj A.M

COURSE OUTCOME COVERED

CO5: Students will be able to analyze and explain advanced generative deep learning models including Autoencoders, Deep Belief Networks, Boltzmann Machines, Deep Boltzmann Machines and Generative Adversarial Networks. They will be able to compare their architectures, explain their learning mechanisms, and apply suitable generative models to real-world problems. (BTL-3)

Course Outcome Weightage: 35%
Assessment Tool Weightage: 50%

Duration: 90 minutes
Mark : 25

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Name of the student:

Criteria	Understanding of Problem Context (2)	Application of Concepts (3)	Problem-Solving Approach (3)	Accuracy of Solution (2)	Clarity (1)	Total (10)
Weightage	20 %	30%	30%	20 %	10 %	100%
Q1						
Total						

Signature of the Faculty:

Total Marks:

PHOTO :



QUESTIONS:

4. Deep Boltzmann Machines

Prepare a technical presentation on Deep Boltzmann Machines, explaining their multilayer

architecture, learning mechanism, interaction among visible and hidden layers, and advantages over conventional Boltzmann Machines. Discuss a practical application and justify why a Deep Boltzmann Machine is appropriate for the selected problem.

PRESENTATION ANSWER :

Technical Presentation: Deep Boltzmann Machines (DBM)

Introduction

Deep Boltzmann Machine (DBM) is a probabilistic generative neural network with multiple hidden layers. It learns complex patterns by modeling the probability distribution of input data.

Key points:

- Unsupervised learning
- Multiple hidden layers
- Learns hierarchical features

- Generates meaningful data representations

DBM Architecture

Layers:

- Visible Layer: Receives input data
- Hidden Layer 1: Learns low-level features
- Hidden Layer 2: Learns high-level representations

Learning Mechanism

DBM learns by minimizing the energy function of the network.

Learning steps:

1. Input is given to visible layer.
2. Hidden neurons are activated.
3. Gibbs Sampling estimates probabilities.
4. Weights are updated to reduce energy.

Interaction Between Layers

Interaction:

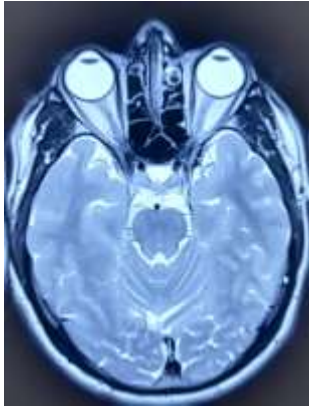
- Visible ↔ Hidden 1
- Hidden 1 ↔ Hidden 2
- Information flows in both directions during inference.
- No connections exist within the same layer.

DBM vs Conventional Boltzmann Machine

Feature	Boltzmann Machine	Deep Boltzmann Machine
Hidden layers	One	Multiple
Feature learning	Simple	Hierarchical
Representation	Low-level	High-level
Accuracy	Moderate	Higher
Complexity	Low	High

DBM learns deeper and richer representations than conventional Boltzmann Machines.

Practical Application – Medical Image Analysis



Problem

Detect diseases from medical images using limited labeled data.

Why DBM?

- Learns hidden image features automatically.
- Reduces manual feature engineering.
- Improves disease classification accuracy.
- Handles uncertainty in medical data.

Why DBM is Suitable?

Advantages:

- Hierarchical feature learning
- Better generative capability
- Robust probabilistic inference
- Improved representation of complex data
- Useful for images, speech, and biomedical applications

Justification: Medical images contain complex patterns. DBM captures both low-level textures and high-level anatomical structures, making it appropriate for disease detection.

Conclusion

- DBM is a deep probabilistic generative model.
- Uses visible and multiple hidden layers.
- Trained using Gibbs Sampling and energy minimization.
- Provides better feature learning than conventional Boltzmann Machines.
- Widely applicable in healthcare, vision, speech, and recommendation systems.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 Exemplary	Level 3 Proficient	Level 2 Developing	Level 1 Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand